

COMPASS_Synoptic_SEAL_Data_Analysis_July2022

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##Run notes

```
Run_notes = "Several GCrew samples were duplicated without a dilution for Vanadium NOx, appears to be in  
Subsetted dataframe in this code to remove samples run orginally and keep the reruns."
```

Information

```
#####  
##### COMPASS Synoptic  
##### Chesapeake Bay Sites  
##### Data Analysis Code: Porewater Nutrients  
##### MONTH: August 2022  
#####  
  
##### Information #####  
#Author: Stephanie J. Wilson  
#Edited: 20220727  
  
#Samples taken from Lysimeters & Sippers at CB Synoptic Sites  
#Samples were filtered with 0.45 uM, kept on ice, frozen until analysis  
#Field Protocol:  
#Samples Analyzed on a SEAL discrete auto analyzer  
#Lab Protocol: https://docs.google.com/document/d/1VaJT7Wb9AcdmM1tgsR\_9ZtQ6kwcaoNmp/edit?usp=sharing&ou  
#NOx method = https://drive.google.com/file/d/1sicqBFnzVxmDd5I2\_pu8s8pj7iNOAuhF/view?usp=sharing  
#NH4 method = https://drive.google.com/file/d/1ENGemUEvm\_rffZqv3lz9BjD0pAMX5nzu/view?usp=sharing  
#PO4 method = https://drive.google.com/file/d/1m3gXDZnJoIo\_QmyhvuZG4HRgGShCzm9Wq/view?usp=sharing  
#Units from SEAL = mg/L and converted to uMoles/L  
  
#QAQC  
#R2 and Slopes  
#These are checked for drift or differences in this code  
  
#Duplicates and Spikes  
#This is checked by the SEAL software  
#If 80% of the dups and spikes are within range we accept  
#the only exceptions are NOx values that are below the detection limit - we do not count dups out of range  
#these are run roughly every 10-12 samples per tray.  
  
#CCV and CCBs
```

```

#CCV's are 50% of the top standard checks run every 10 samples, these are checked by the software; if 80%
#CCB's are DI blanks run every 10 samples, these are checked by the software; if 80% are within range w

#Third Party Standard
#This is checked within the code - must be within 20% of the peChk concentration

```

QAQC on Slopes

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
library(data.table)
```

```

#read in datafile with all the slopes
qlog <- read.csv("Raw Data/SERC_SEAL_STDs_Log.csv")
head(qlog)

```

```

##      Date      User  Machine Analysis  Slope Intercept    R2
## 1 20220513 Stephanie Wilson SERC SEAL    NH3 1.9370   -0.030 0.9995
## 2 20220513 Stephanie Wilson SERC SEAL    P04 2.8690    0.002 0.9994
## 3 20220525 Stephanie Wilson SERC SEAL  V-Nox 1.0000    0.000 0.9990
## 4 20220614 Stephanie Wilson SERC SEAL    NH3 1.1487   -0.011 0.9999
## 5 20220614 Stephanie Wilson SERC SEAL    P04 2.3950   -0.001 0.9998
## 6 20220615 Stephanie Wilson SERC SEAL  V-Nox 1.0000    0.000 0.9998
##   Nox_Red_Eff X
## 1             NA
## 2             NA
## 3             NA
## 4             NA
## 5             NA
## 6             NA

```

```

#pull out each method
qNH3 <- qlog[qlog$Analysis %like% "NH3", ]
head(qNH3)

```

```

##      Date      User  Machine Analysis  Slope Intercept    R2
## 1 20220513 Stephanie Wilson SERC SEAL    NH3 1.9370   -0.030 0.9995
## 4 20220614 Stephanie Wilson SERC SEAL    NH3 1.1487   -0.011 0.9999
## 7 20220615 Stephanie Wilson SERC SEAL    NH3 1.7070   -0.008 1.0000
## 10 20220718 Stephanie Wilson SERC SEAL    NH3 1.7080   -0.017 0.9995
## 12 20220718 Stephanie Wilson SERC SEAL    NH3 1.6590   -0.017 0.9997
## 14 20220719 Stephanie Wilson SERC SEAL    NH3 1.5560    0.000 0.9994
##   Nox_Red_Eff X
## 1             NA
## 4             NA
## 7             NA
## 10            NA
## 12            NA
## 14            NA

```

```
qP04 <- qlog[qlog$Analysis %like% "P04", ]
head(qP04)
```

```
##      Date      User  Machine Analysis Slope Intercept      R2
## 2  20220513 Stephanie Wilson  SERC SEAL      P04  2.869      0.002 0.9994
## 5  20220614 Stephanie Wilson  SERC SEAL      P04  2.395     -0.001 0.9998
## 8  20220615 Stephanie Wilson  SERC SEAL      P04  2.349      0.000 0.9999
## 11 20220718 Stephanie Wilson  SERC SEAL      P04  2.445     -0.003 0.9993
## 13 20220718 Stephanie Wilson  SERC SEAL      P04  2.349     -0.004 0.9992
## 15 20220719 Stephanie Wilson  SERC SEAL      P04  2.271     -0.003 0.9993
##      Nox_Red_Eff X
## 2              NA
## 5              NA
## 8              NA
## 11             NA
## 13             NA
## 15             NA
```

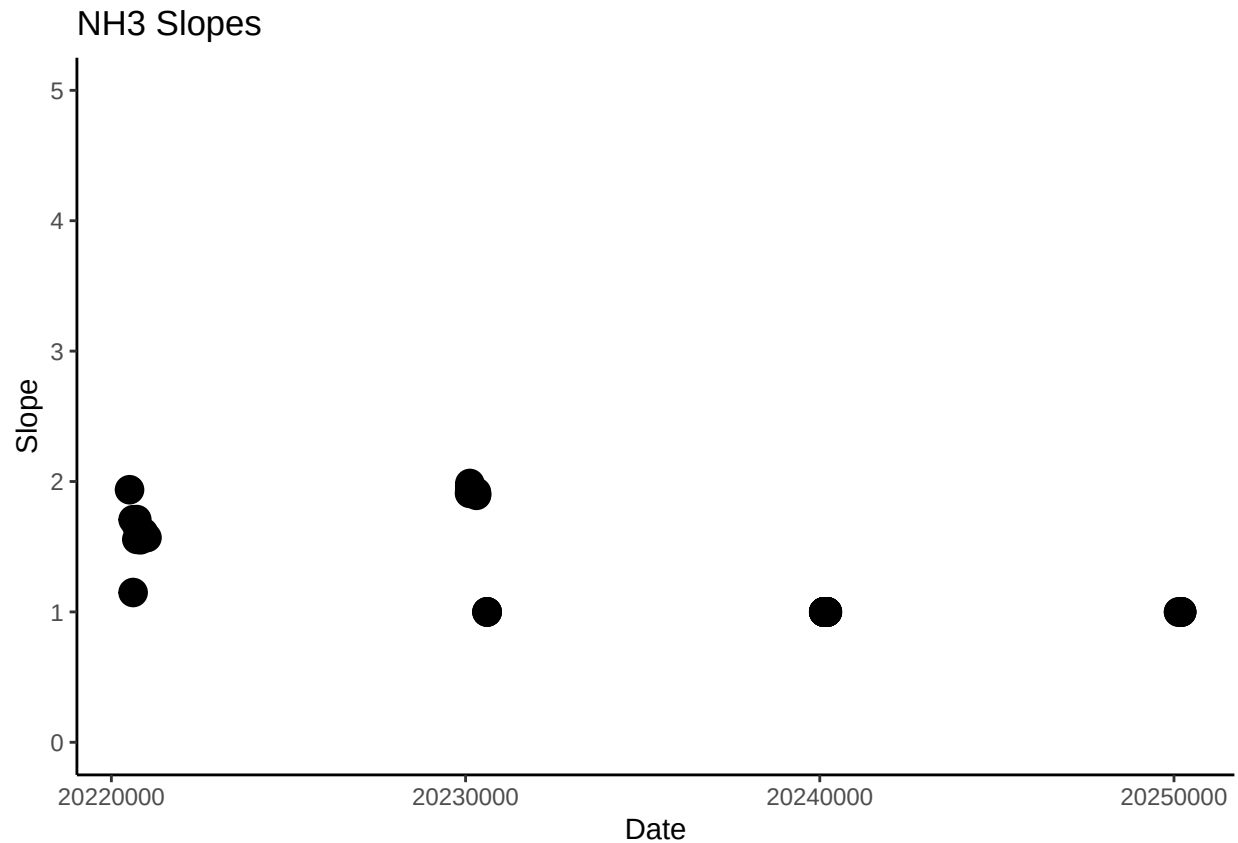
```
qNOx <- qlog[qlog$Analysis %like% "V-Nox", ]
head(qNOx)
```

```
##      Date      User  Machine Analysis Slope Intercept      R2
## 3  20220525 Stephanie Wilson  SERC SEAL      V-Nox      1          0 0.9990
## 6  20220615 Stephanie Wilson  SERC SEAL      V-Nox      1          0 0.9998
## 9  20220718 Stephanie Wilson  SERC SEAL      V-Nox      1          0 0.9999
## 16 20220730 Stephanie Wilson  SERC SEAL      V-Nox      1          0 1.0000
## 17 20220731 Stephanie Wilson  SERC SEAL      V-Nox      1          0 1.0000
## 22 20220824 Stephanie Wilson  SERC SEAL      V-Nox      1          0 0.9988
##      Nox_Red_Eff X
## 3              NA
## 6              NA
## 9              NA
## 16             NA
## 17             NA
## 22             NA
```

```
##### NH3
#plot the slopes to make sure there are no crazy outliers
slope1 <- ggplot(data=qNH3, aes(x=Date, y=Slope)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0, 5) +
  theme(legend.position="none") +
  ggtitle("NH3 Slopes")

slope1
```

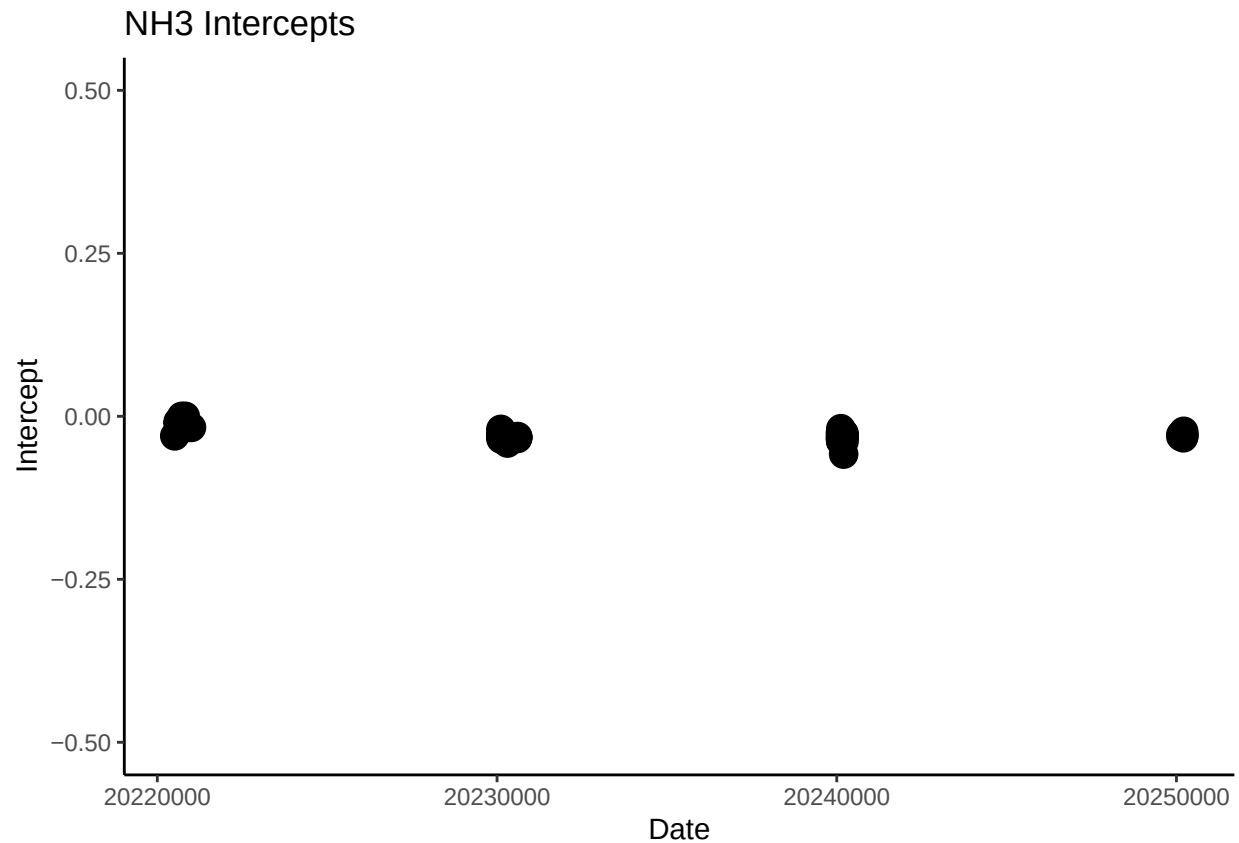
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the intercepts to make sure there are no crazy outliers
int1 <- ggplot(data=qNH3, aes(x=Date, y=Intercept)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(-0.5,0.5) +
  theme(legend.position="none")+
  ggtitle("NH3 Intercepts")

int1
```

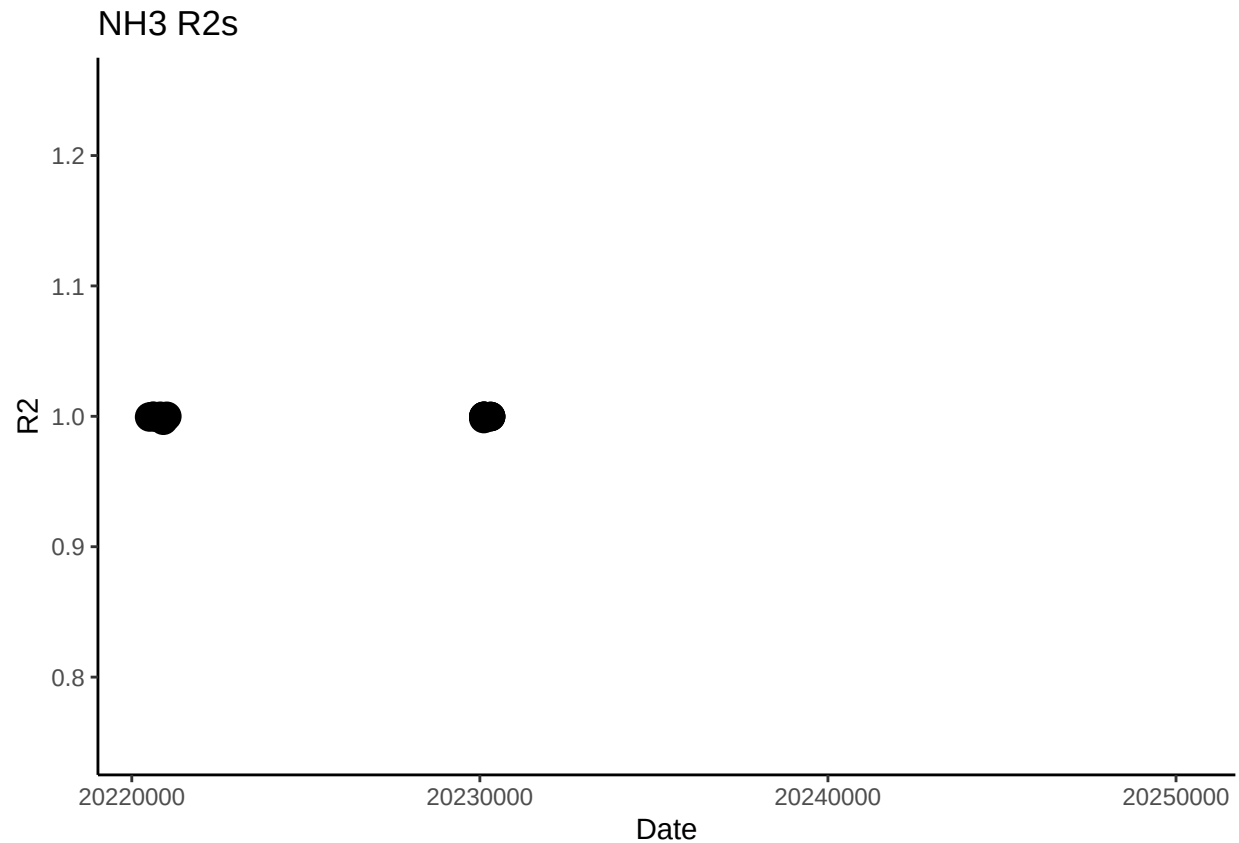
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the R2s to make sure there are no crazy outliers
Rsqr1 <- ggplot(data=qNH3, aes(x=Date, y=R2)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0.75, 1.25) +
  theme(legend.position="none")+
  ggtitle("NH3 R2s")
```

Rsqr1

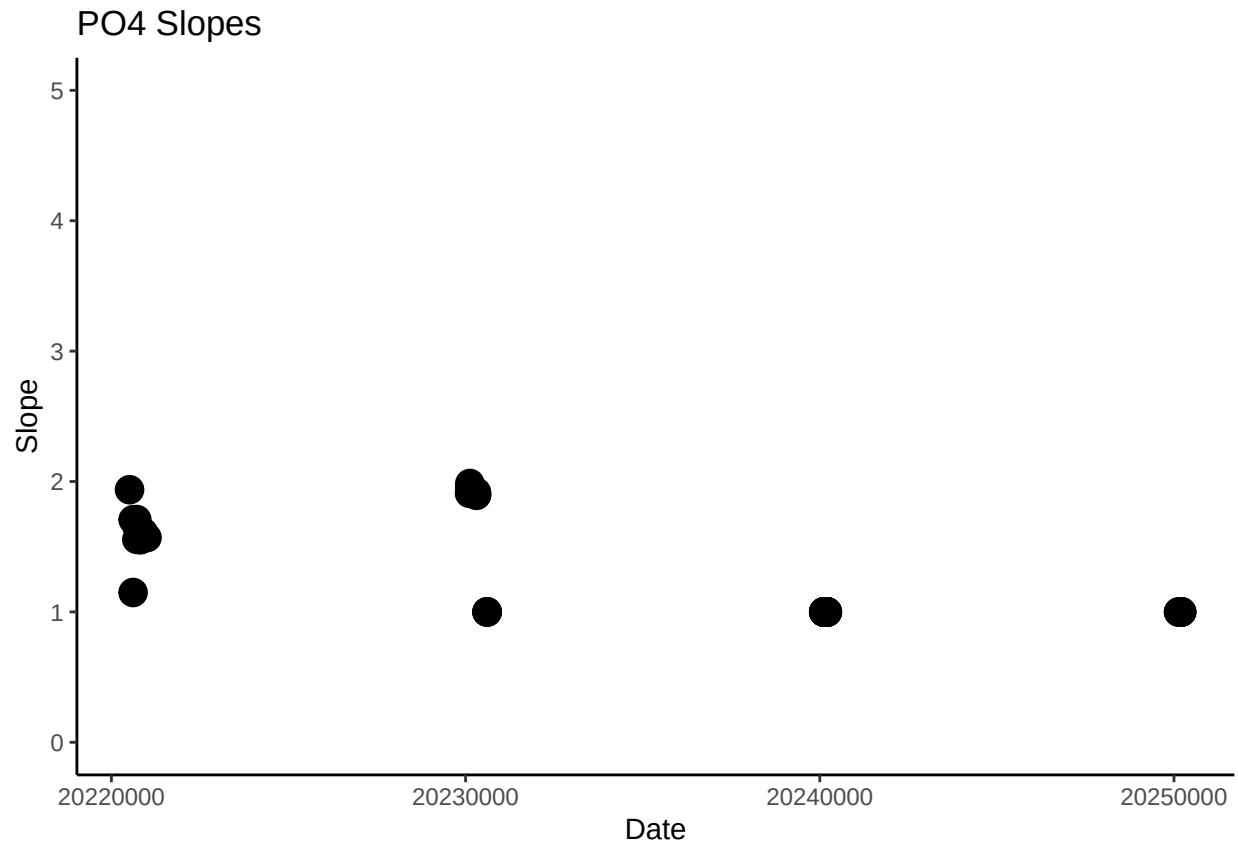
```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
##### P04
slope2 <- ggplot(data=qNH3, aes(x=Date, y=Slope)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0, 5) +
  theme(legend.position="none")+
  ggtitle("P04 Slopes")

slope2
```

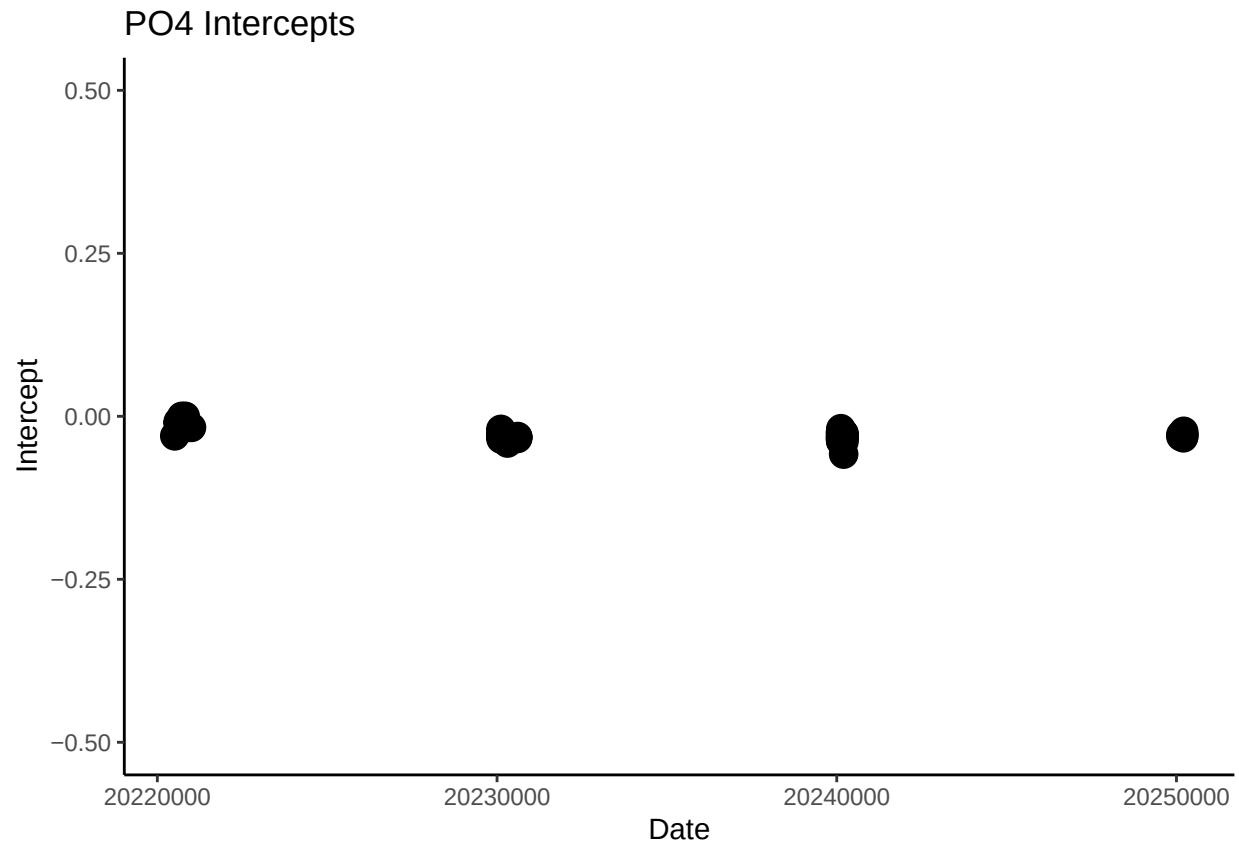
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the intercepts to make sure there are no crazy outliers
int2 <- ggplot(data=qNH3, aes(x=Date, y=Intercept)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(-0.5,0.5) +
  theme(legend.position="none")+
  ggtitle("P04 Intercepts")

int2
```

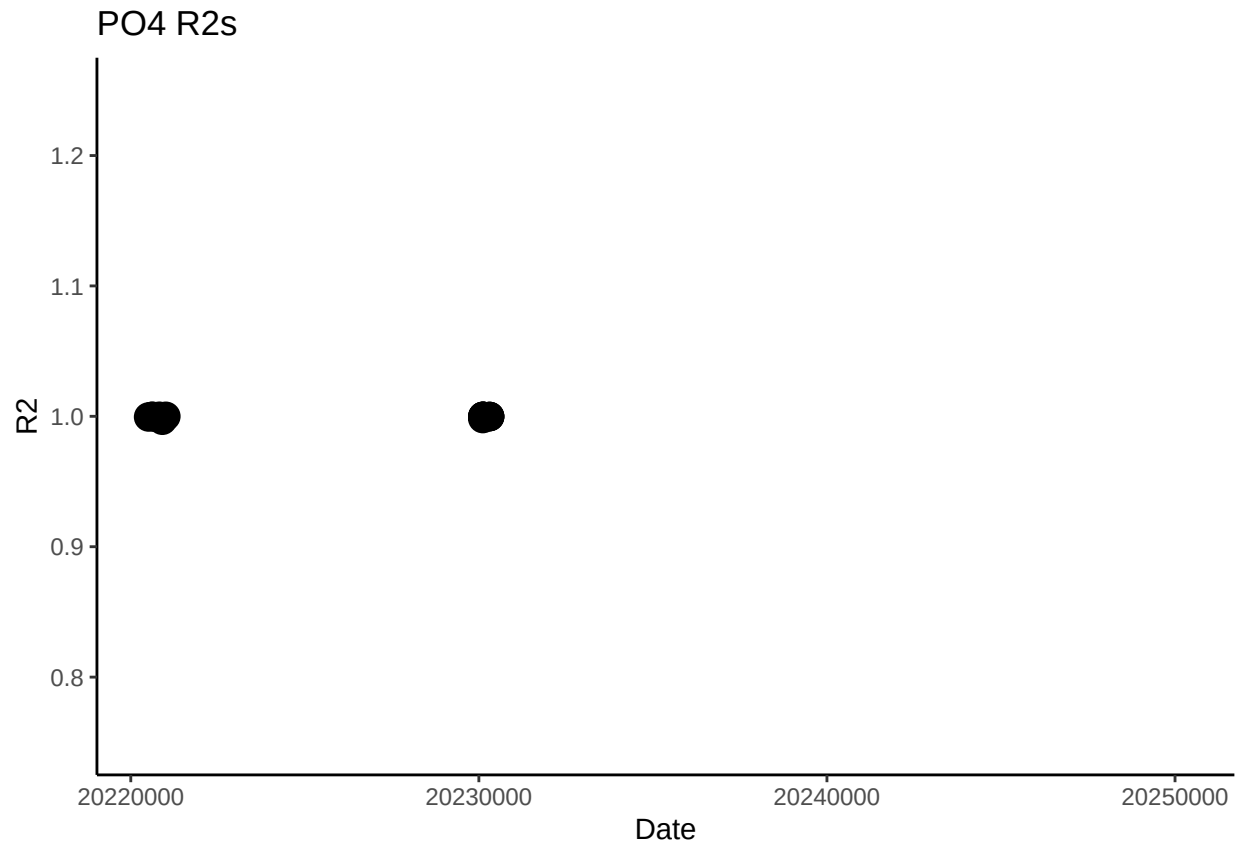
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the R2s to make sure there are no crazy outliers
Rsqr2 <- ggplot(data=qNH3, aes(x=Date, y=R2)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0.75, 1.25) +
  theme(legend.position="none")+
  ggtitle("PO4 R2s")
```

Rsqr2

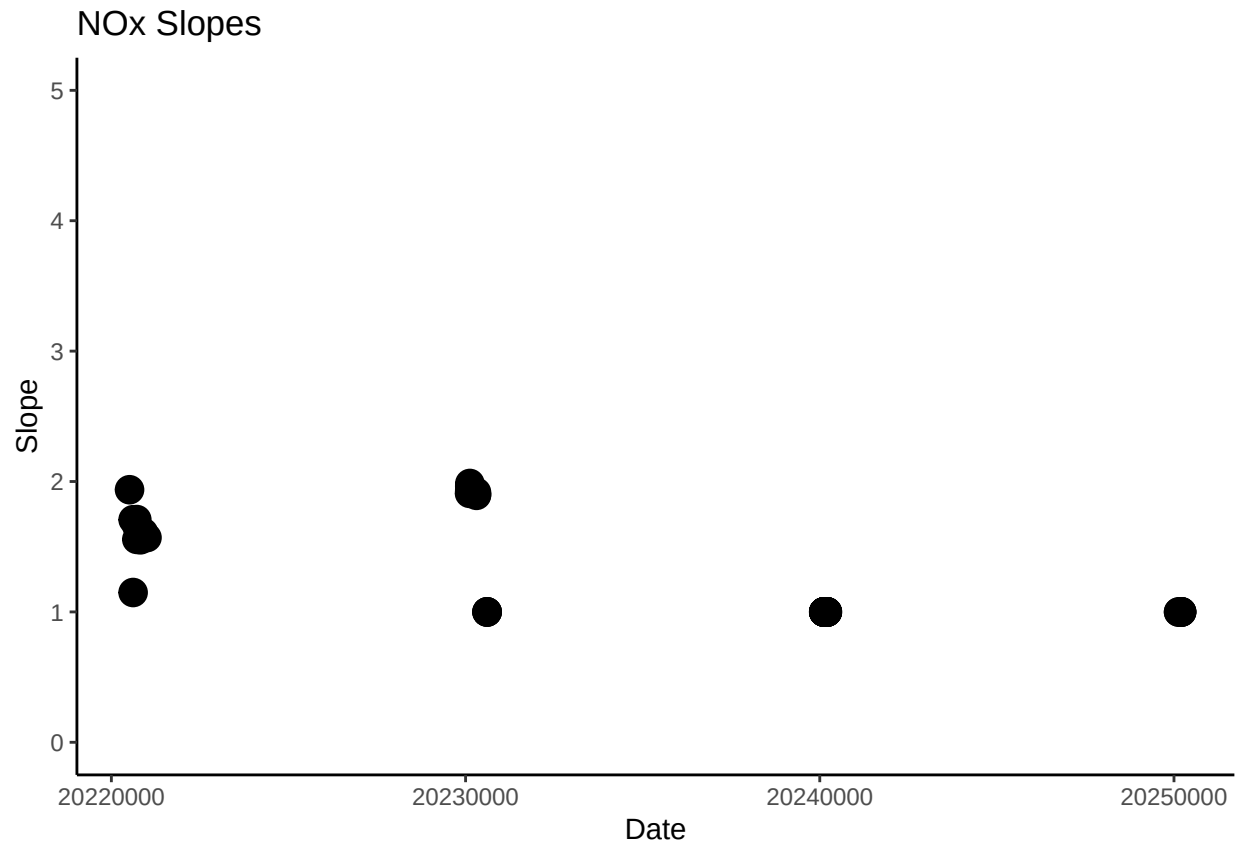
```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
##### NOx
slope3 <- ggplot(data=qNH3, aes(x=Date, y=Slope)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0, 5) +
  theme(legend.position="none")+
  ggtitle("NOx Slopes")

slope3
```

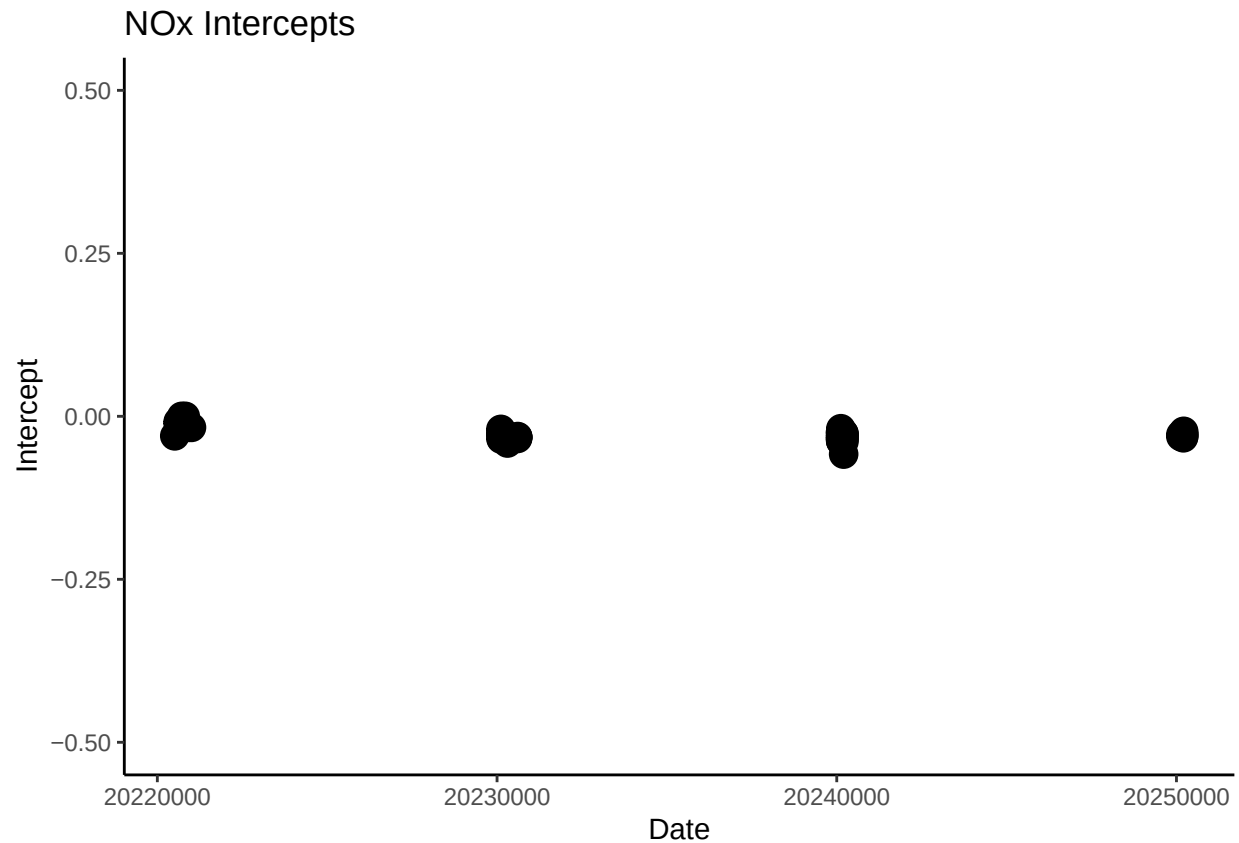
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the intercepts to make sure there are no crazy outliers
int3 <- ggplot(data=qNH3, aes(x=Date, y=Intercept)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(-0.5,0.5) +
  theme(legend.position="none")+
  ggtitle("NOx Intercepts")

int3
```

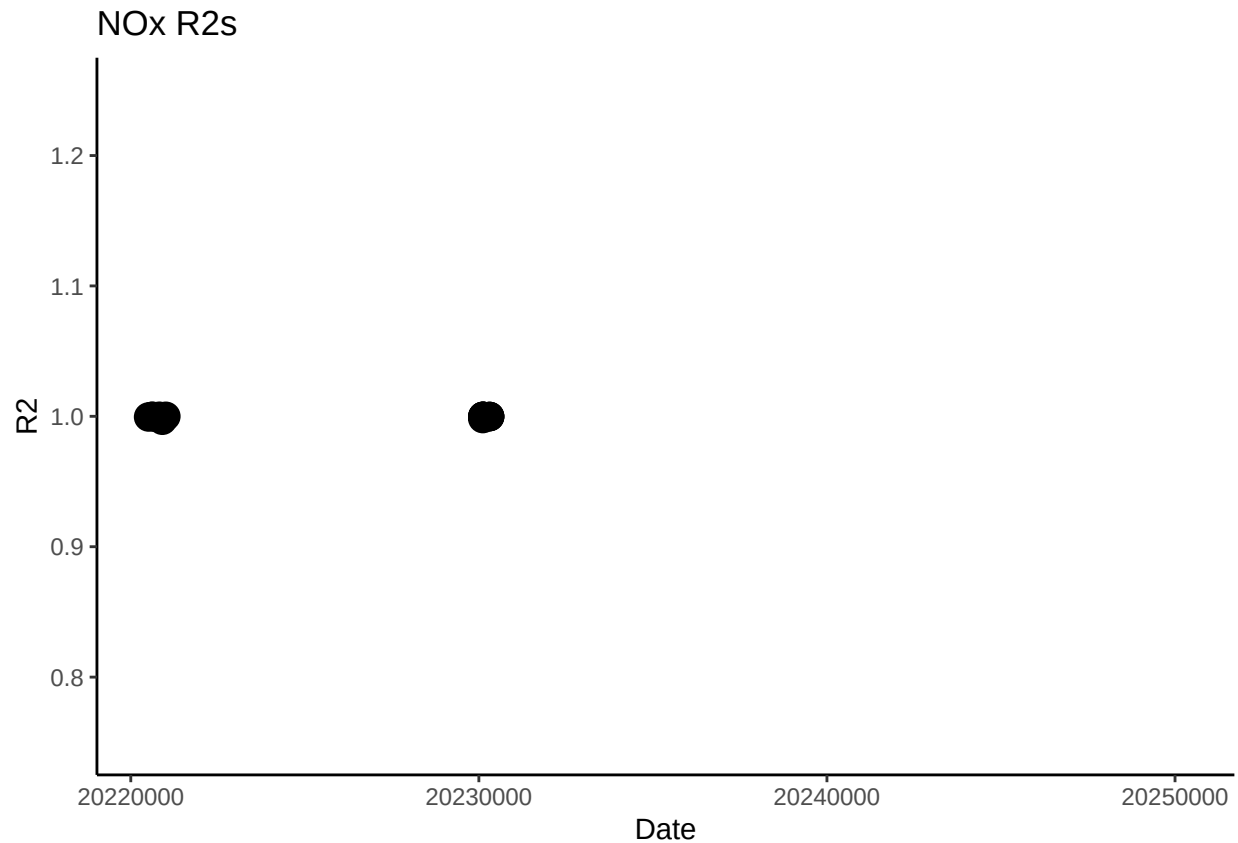
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the R2s to make sure there are no crazy outliers
Rsqr3 <- ggplot(data=qNH3, aes(x=Date, y=R2)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0.75, 1.25) +
  theme(legend.position="none")+
  ggtitle("NOx R2s")
```

Rsqr3

```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Code Set up

```
#packages:
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:data.table':
##
##   between, first, last

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(data.table)
library(matrixStats)
```

```
##
## Attaching package: 'matrixStats'

## The following object is masked from 'package:dplyr':
##
## count
```

```
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
## combine
```

```
library(ggpubr)
library(grid)
```

Ammonia & Phosphate

```
#read in data
file1 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202207_1.csv")
#Quick look at dataframe
head(file1)
```

```
## RUNSTARTED X1658165226 X7.18.2022.13.27 X X.1 Conc Abs
## 1 QCPROACTION CONTINUE 1658165413 NA
## 2 RUNENDED 1658165413 COMPLETE NA
## 3 RUNSTARTED 1658165922 7/18/2022 13:38 NA
## 4 RESULT -1 S1 Standard 1 0 0.006887 0.006887
## 5 RESULT -2 S90 Standard .0389 1 0.03206 0.03206
## 6 RESULT -2 S91 Standard .1000 2 0.065922 0.065922
## X.2 Dilution X.3 X.4 X.5 X.6 X.7 X.8
## 1 NA NA NA NA NA
## 2 NA NA NA NA NA
## 3 NA NA NA NA NA
## 4 0 0 0 0 mg N/L Ammonia 2 1658167154 7/18/2022 13:59
## 5 0 0 0 0 mg N/L Ammonia 2 1658167314 7/18/2022 14:01
## 6 0 0 0 0 mg N/L Ammonia 2 1658167474 7/18/2022 14:04
```

```
#take out only the columns that we need
dat1 <- file1[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(dat1) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
dat1$Run_Number <- "1"
head(dat1)
```

```
##      Run_Info      Sample_Name      Conc      Abs Dilution  Units      Test
## 1 QCPROACTION
## 2      RUNENDED
## 3 RUNSTARTED
## 4      RESULT      Standard 1 0.006887 0.006887      0 mg N/L Ammonia 2
## 5      RESULT Standard .0389 0.03206 0.03206      0 mg N/L Ammonia 2
## 6      RESULT Standard .1000 0.065922 0.065922      0 mg N/L Ammonia 2
##      Run_Number
## 1          1
## 2          1
## 3          1
## 4          1
## 5          1
## 6          1
```

```
#read in data
file2 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202207_2.csv")
#Quick look at dataframe
head(file2)
```

```
##      RUNSTARTED X1658190607      X7.18.2022.20.30      X X.1      X.2
## 1      RUNENDED 1658190628 STOP IMMEDIATE PRESSED      NA
## 2 RUNSTARTED 1658190693      7/18/2022 20:31      NA
## 3
## 4      RESULT      -1      S1      Standard 1 0 0.005962
## 5      RESULT      -2      S90 Standard .0389 1 0.030602
## 6      RESULT      -2      S91 Standard .1000 2 0.066232
##      X.3 X.4 X.5 X.6 X.7      X.8      X.9      X.10      X.11
## 1      NA      NA      NA      NA
## 2      NA      NA      NA      NA
## 3      Abs NA Dil NA NA      NA
## 4 0.005962 0 0 0 512 mg N/L Ammonia 2 1658191931 7/18/2022 20:52
## 5 0.030602 0 0 0 512 mg N/L Ammonia 2 1658192091 7/18/2022 20:54
## 6 0.066232 0 0 0 512 mg N/L Ammonia 2 1658192251 7/18/2022 20:57
```

```
#take out only the columns that we need
dat2 <- file2[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(dat2) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
dat2$Run_Number <- "2"
head(dat2)
```

```
##      Run_Info      Sample_Name      Conc      Abs Dilution  Units      Test
## 1      RUNENDED
## 2 RUNSTARTED
## 3
## 4      RESULT      Standard 1 0.005962 0.005962      0 mg N/L Ammonia 2
## 5      RESULT Standard .0389 0.030602 0.030602      0 mg N/L Ammonia 2
## 6      RESULT Standard .1000 0.066232 0.066232      0 mg N/L Ammonia 2
##      Run_Number
## 1          2
## 2          2
## 3          2
```

```
## 4      2
## 5      2
## 6      2
```

```
#read in data
file3 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202207_3.csv")
#Quick look at dataframe
head(file3)
```

```
##  RUNSTARTED X1658228831      X7.19.2022.7.07      X X.1      X.2
## 1 RUNSTARTED 1658230325      7/19/2022 7:32      NA      NA
## 2 RUNENDED 1658230365 STOP IMMEDIATE PRESSED      NA      NA
## 3 RUNSTARTED 1658230665      7/19/2022 7:37      NA      NA
## 4 RESULT      -1      S1      Standard 1 0 0.005266
## 5 RESULT      -2      S90 Standard .0389 1 0.031630
## 6 RESULT      -2      S91 Standard .1000 2 0.067535
##      X.3 X.4 X.5 X.6 X.7      X.8      X.9      X.10      X.11
## 1      NA NA NA NA NA      NA      NA      NA
## 2      NA NA NA NA NA      NA      NA      NA
## 3      NA NA NA NA NA      NA      NA      NA
## 4 0.005266 0 0 0 0 mg N/L Ammonia 2 1658231899 7/19/2022 7:58
## 5 0.031630 0 0 0 0 mg N/L Ammonia 2 1658232059 7/19/2022 8:00
## 6 0.067535 0 0 0 0 mg N/L Ammonia 2 1658232219 7/19/2022 8:03
```

```
#take out only the columns that we need
dat3 <- file3[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(dat3) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
dat3$Run_Number <- "3"
head(dat3)
```

```
##  Run_Info      Sample_Name      Conc      Abs Dilution Units      Test
## 1 RUNSTARTED      NA      NA      NA
## 2 RUNENDED      NA      NA      NA
## 3 RUNSTARTED      NA      NA      NA
## 4 RESULT      Standard 1 0.005266 0.005266      0 mg N/L Ammonia 2
## 5 RESULT Standard .0389 0.031630 0.031630      0 mg N/L Ammonia 2
## 6 RESULT Standard .1000 0.067535 0.067535      0 mg N/L Ammonia 2
##  Run_Number
## 1      3
## 2      3
## 3      3
## 4      3
## 5      3
## 6      3
```

```
alldat <- rbind(dat1, dat2, dat3)

#Pull out standards
stds <- alldat[alldat$Sample_Name %like% "Standard", ]
head(stds)
```

```
##  Run_Info      Sample_Name      Conc      Abs Dilution Units      Test
```

```
## 4   RESULT      Standard 1 0.006887 0.006887      0 mg N/L Ammonia 2
## 5   RESULT Standard .0389  0.03206  0.03206      0 mg N/L Ammonia 2
## 6   RESULT Standard .1000  0.065922 0.065922      0 mg N/L Ammonia 2
## 7   RESULT Standard .2000  0.130039 0.130039      0 mg N/L Ammonia 2
## 8   RESULT Standard .5000   0.3005   0.3005      0 mg N/L Ammonia 2
## 9   RESULT Standard 1.0000 0.618677 0.618677      0 mg N/L Ammonia 2
##   Run_Number
## 4           1
## 5           1
## 6           1
## 7           1
## 8           1
## 9           1
```

```
#Pull out samples
```

```
alldat2 <- alldat[alldat$Sample_Name %like% "MSM_", ]
alldat2 <- rbind(alldat2, (alldat[alldat$Sample_Name %like% "GWI_", ]))
alldat2 <- rbind(alldat2, (alldat[alldat$Sample_Name %like% "GCrew_", ]))
head(alldat2)
```

```
##   Run_Info      Sample_Name      Conc      Abs Dilution  Units      Test
## 398  RESULT MSM_UP_LysB_10cm 0.042758 0.035969      0 mg N/L Ammonia 2
## 405  RESULT MSM_UP_LysC_10cm 0.568162 0.352609      0 mg N/L Ammonia 2
## 406  RESULT MSM_UP_LysA_20cm 1.795753 1.092427      0 mg N/L Ammonia 2
## 414  RESULT MSM_UP_LysB_20cm 0.588825 0.365062      0 mg N/L Ammonia 2
## 415  RESULT MSM_UP_LysC_20cm 1.999664 1.215316      0 mg N/L Ammonia 2
## 418  RESULT MSM_UP_LysA_45cm 1.148694 0.702472      0 mg N/L Ammonia 2
##   Run_Number
## 398         2
## 405         2
## 406         2
## 414         2
## 415         2
## 418         2
```

```
alldat3 <- alldat2
```

NO_x

```
#read in data
```

```
Nfile1 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NOx_202207_1.csv")
```

```
#Quick look at dataframe
```

```
head(Nfile1)
```

```
##   RUNSTARTED X1658149249 X7.18.2022.9.00      X X.1      Conc      Abs X.2
## 1   RESULT          -1          S1 Standard 1    0 0.165420 0.165420    0
## 2   RESULT          -2          S90 Standard 90   1 0.182766 0.182766    0
## 3   RESULT          -2          S91 Standard 91   2 0.207313 0.207313    0
## 4   RESULT          -2          S92 Standard 92   3 0.248635 0.248635    0
## 5   RESULT          -2          S93 Standard 93   4 0.360514 0.360514    0
## 6   RESULT          -2          S94 Standard 94   5 0.563482 0.563482    0
```



```
##   Dil X.3 X.4   X.5           X.6           X.7           X.8
## 1   0   0   0 mg N/L Vanadium NOx 1658151476 7/18/2022 9:37
## 2   0   0   0 mg N/L Vanadium NOx 1658151558 7/18/2022 9:39
## 3   0   0   0 mg N/L Vanadium NOx 1658151639 7/18/2022 9:40
## 4   0   0   0 mg N/L Vanadium NOx 1658151720 7/18/2022 9:42
## 5   0   0   0 mg N/L Vanadium NOx 1658151800 7/18/2022 9:43
## 6   0   0   0 mg N/L Vanadium NOx 1658151882 7/18/2022 9:44
```

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:data.table':
##
##   hour, isoweek, mday, minute, month, quarter, second, wday, week,
##   yday, year

## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

```
#This run was stopped and restarted... need to take out the rows that are not needed:
Nfile1 <- Nfile1 %>%
  mutate(X.8 = mdy_hm(X.8)) %>%           # parse timestamp
  filter(format(X.8, "%H:%M") >= "11:26")

#take out only the columns that we need
Ndat1 <- Nfile1[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(Ndat1) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
Ndat1$Run_Number <- "1"
head(Ndat1)
```

```
##   Run_Info      Sample_Name      Conc      Abs Dilution Units      Test
## 1  RESULT      Standard 96 0.948336 0.948336      0 mg N/L Vanadium NOx
## 2  RESULT      Standard 0 0.169071 0.169071      0 mg N/L Vanadium NOx
## 3  RESULT Nitrate Standard 0.499425 0.575431      0 mg N/L Vanadium NOx
## 4  RESULT Nitrite Standard 0.461612 0.545854      0 mg N/L Vanadium NOx
## 5  RESULT              CCV 0.487806 0.566363      0 mg N/L Vanadium NOx
## 6  RESULT              CCB 0.002467 0.169996      0 mg N/L Vanadium NOx
##   Run_Number
## 1           1
## 2           1
## 3           1
## 4           1
## 5           1
## 6           1
```

```
#read in data
Nfile2 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NOx_202207_2.csv")
#Quick look at dataframe
head(Nfile2)
```

```
## RUNSTARTED X1659202700 X7.30.2022.13.38 X X.1 X.2 X.3 X.4
## 1 RUNSTARTED 1659202912 7/30/2022 13:41 NA NA NA NA
## 2 RESULT -1 S1 Standard 1 0 0.165839 0.165839 0
## 3 RESULT -2 S90 Standard 90 1 0.178468 0.178468 0
## 4 RESULT -2 S91 Standard 91 2 0.202958 0.202958 0
## 5 RESULT -2 S92 Standard 92 3 0.240962 0.240962 0
## 6 RESULT -2 S93 Standard 93 4 0.359596 0.359596 0
## X.5 X.6 X.7 X.8 X.9 X.10 X.11
## 1 NA NA NA NA
## 2 0 0 0 mg N/L Vanadium NOx 1659205111 7/30/2022 14:18
## 3 0 0 0 mg N/L Vanadium NOx 1659205193 7/30/2022 14:19
## 4 0 0 0 mg N/L Vanadium NOx 1659205274 7/30/2022 14:21
## 5 0 0 0 mg N/L Vanadium NOx 1659205354 7/30/2022 14:22
## 6 0 0 0 mg N/L Vanadium NOx 1659205435 7/30/2022 14:23
```

```
#take out only the columns that we need
Ndat2 <- Nfile2[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(Ndat2) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
Ndat2$Run_Number <- "2"
head(Ndat2)
```

```
## Run_Info Sample_Name Conc Abs Dilution Units Test
## 1 RUNSTARTED NA NA NA
## 2 RESULT Standard 1 0.165839 0.165839 0 mg N/L Vanadium NOx
## 3 RESULT Standard 90 0.178468 0.178468 0 mg N/L Vanadium NOx
## 4 RESULT Standard 91 0.202958 0.202958 0 mg N/L Vanadium NOx
## 5 RESULT Standard 92 0.240962 0.240962 0 mg N/L Vanadium NOx
## 6 RESULT Standard 93 0.359596 0.359596 0 mg N/L Vanadium NOx
## Run_Number
## 1 2
## 2 2
## 3 2
## 4 2
## 5 2
## 6 2
```

```
#read in data
Nfile3 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NOx_202207_3.csv")
#Quick look at dataframe
head(Nfile3)
```

```
## RUNSTARTED X1659298379 X7.31.2022.16.12 X X.1 X.2 X.3 X.4
## 1 RESULT -1 S1 Standard 1 0 0.172549 0.172549 0
## 2 RESULT -2 S90 Standard 90 1 0.194234 0.194234 0
## 3 RESULT -2 S91 Standard 91 2 0.214575 0.214575 0
## 4 RESULT -2 S92 Standard 92 3 0.253198 0.253198 0
## 5 RESULT -2 S93 Standard 93 4 0.377676 0.377676 0
## 6 RESULT -2 S94 Standard 94 5 0.585169 0.585169 0
## X.5 X.6 X.7 X.8 X.9 X.10 X.11
## 1 0 0 0 mg N/L Vanadium NOx 1659300574 7/31/2022 16:49
## 2 0 0 0 mg N/L Vanadium NOx 1659300655 7/31/2022 16:50
## 3 0 0 0 mg N/L Vanadium NOx 1659300737 7/31/2022 16:52
```

```
## 4 0 0 0 mg N/L Vanadium NOx 1659300817 7/31/2022 16:53
## 5 0 0 0 mg N/L Vanadium NOx 1659300899 7/31/2022 16:54
## 6 0 0 0 mg N/L Vanadium NOx 1659300980 7/31/2022 16:56
```

```
#take out only the columns that we need
Ndat3 <- Nfile3[ ,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(Ndat3) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
Ndat3$Run_Number <- "3"
head(Ndat3)
```

```
## Run_Info Sample_Name Conc Abs Dilution Units Test
## 1 RESULT Standard 1 0.172549 0.172549 0 mg N/L Vanadium NOx
## 2 RESULT Standard 90 0.194234 0.194234 0 mg N/L Vanadium NOx
## 3 RESULT Standard 91 0.214575 0.214575 0 mg N/L Vanadium NOx
## 4 RESULT Standard 92 0.253198 0.253198 0 mg N/L Vanadium NOx
## 5 RESULT Standard 93 0.377676 0.377676 0 mg N/L Vanadium NOx
## 6 RESULT Standard 94 0.585169 0.585169 0 mg N/L Vanadium NOx
## Run_Number
## 1 3
## 2 3
## 3 3
## 4 3
## 5 3
## 6 3
```

```
Nalldat <- rbind(Ndat1, Ndat2, Ndat3)
```

```
#Pull out standards
Nstds <- Nalldat[Nalldat$Sample_Name %like% "Standard", ]
head(Nstds)
```

```
## Run_Info Sample_Name Conc Abs Dilution Units Test
## 1 RESULT Standard 96 0.948336 0.948336 0 mg N/L Vanadium NOx
## 2 RESULT Standard 0 0.169071 0.169071 0 mg N/L Vanadium NOx
## 3 RESULT Nitrate Standard 0.499425 0.575431 0 mg N/L Vanadium NOx
## 4 RESULT Nitrite Standard 0.461612 0.545854 0 mg N/L Vanadium NOx
## 72 RESULT Standard 1 0.165839 0.165839 0 mg N/L Vanadium NOx
## 73 RESULT Standard 90 0.178468 0.178468 0 mg N/L Vanadium NOx
## Run_Number
## 1 1
## 2 1
## 3 1
## 4 1
## 72 2
## 73 2
```

```
#Pull out samples
Nalldat2 <- Nalldat[Nalldat$Sample_Name %like% "MSM_", ]
Nalldat2 <- rbind(Nalldat2, (Nalldat[Nalldat$Sample_Name %like% "GWI_", ]))
Nalldat2 <- rbind(Nalldat2, (Nalldat[Nalldat$Sample_Name %like% "GCrew_", ]))
head(Nalldat2)
```

```
##      Run_Info      Sample_Name      Conc      Abs Dilution  Units      Test
## 92      RESULT MSM_UP_LysB_10cm 0.001102 0.162520      0 mg N/L Vanadium NOx
## 100     RESULT MSM_UP_LysC_10cm 0.002487 0.163655      0 mg N/L Vanadium NOx
## 101     RESULT MSM_UP_LysA_20cm 0.001262 0.162651      0 mg N/L Vanadium NOx
## 109     RESULT MSM_UP_LysB_20cm 0.003195 0.164234      0 mg N/L Vanadium NOx
## 110     RESULT MSM_UP_LysC_20cm -0.001923 0.160043      0 mg N/L Vanadium NOx
## 111     RESULT MSM_UP_LysA_45cm 0.000066 0.161672      0 mg N/L Vanadium NOx
##      Run_Number
## 92              2
## 100             2
## 101             2
## 109             2
## 110             2
## 111             2
```

```
Nalldat3 <- Nalldat2
```

peCheck Checks

```
#pull out peChecks from alldat and Nalldat
chks <- alldat[alldat$Sample_Name %like% "peCheck", ]
head(chks)
```

```
##      Run_Info      Sample_Name      Conc      Abs Dilution
## 142     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 1.285274 0.762447      0
## 149     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 1.270126 0.753577      0
## 327     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.79617 0.326971      0
## 336     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.7993 0.328251      0
## 376     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.929474 0.077364      5
## 377     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.89814 0.074801      5
##      Units      Test Run_Number
## 142 mg N/L Ammonia 2      1
## 149 mg N/L Ammonia 2      1
## 327 mg P/L o-PHOS 0.3      1
## 336 mg P/L o-PHOS 0.3      1
## 376 mg P/L o-PHOS 0.3      1
## 377 mg P/L o-PHOS 0.3      1
```

```
Nchks <- Nalldat[Nalldat$Sample_Name %like% "peCheck", ]

NH4chk <- subset(chks, Test == "Ammonia 2")
NH4chk$ConcTrue <- 0.948
NH4chk$Conc <- as.numeric(NH4chk$Conc)
NH4chk$Chk_diff <- (NH4chk$Conc - NH4chk$ConcTrue)/((NH4chk$Conc+NH4chk$ConcTrue)/2) * 100
NH4chk$Chk_diff_flag <- ifelse(NH4chk$Chk_diff <25, 'YES', 'NO, rerun')
head(NH4chk)
```

```
##      Run_Info      Sample_Name      Conc      Abs
## 142     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 1.285274 0.762447
## 149     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 1.270126 0.753577
## 571     RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04 1.229047 0.750897
```

```
## 1060 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04_A 1.061506 0.682024
## 1061 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04_B 1.079544 0.693616
##      Dilution Units      Test Run_Number ConcTrue Chk_diff Chk_dff_flag
## 142      0 mg N/L Ammonia 2          1      0.948 30.20444      NO, rerun
## 149      0 mg N/L Ammonia 2          1      0.948 29.04488      NO, rerun
## 571      0 mg N/L Ammonia 2          2      0.948 25.81910      NO, rerun
## 1060      0 mg N/L Ammonia 2          3      0.948 11.29691          YES
## 1061      0 mg N/L Ammonia 2          3      0.948 12.97570          YES
```

```
P04chk <- subset(chks, Test == "o-PHOS 0.3")
P04chk$ConcTrue <- 0.818
P04chk$Conc <- as.numeric(P04chk$Conc)
P04chk$Chk_diff <- (P04chk$Conc - P04chk$ConcTrue)/((P04chk$Conc+P04chk$ConcTrue)/2) * 100
P04chk$Chk_dff_flag <- ifelse(P04chk$Chk_diff <25, 'YES', 'NO, rerun')
head(P04chk)
```

```
##      Run_Info      Sample_Name      Conc      Abs Dilution
## 327 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.796170 0.326971      0
## 336 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.799300 0.328251      0
## 376 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.929474 0.077364      5
## 377 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.898140 0.074801      5
## 800 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04 0.745772 0.319165      0
## 895 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04 0.803470 0.070133      5
##      Units      Test Run_Number ConcTrue Chk_diff Chk_dff_flag
## 327 mg P/L o-PHOS 0.3          1      0.818 -2.704796          YES
## 336 mg P/L o-PHOS 0.3          1      0.818 -2.312496          YES
## 376 mg P/L o-PHOS 0.3          1      0.818 12.758301          YES
## 377 mg P/L o-PHOS 0.3          1      0.818 9.339564          YES
## 800 mg P/L o-PHOS 0.3          2      0.818 -9.237664          YES
## 895 mg P/L o-PHOS 0.3          2      0.818 -1.792201          YES
```

```
NOXchk <- subset(Nchks, Test == "Vanadium NOx")
NOXchk$ConcTrue <- 0.706
NOXchk$Conc <- as.numeric(NOXchk$Conc)
NOXchk$Chk_diff <- (NOXchk$Conc - NOXchk$ConcTrue)/((NOXchk$Conc+NOXchk$ConcTrue)/2) * 100
NOXchk$Chk_dff_flag <- ifelse(NOXchk$Chk_diff <25, 'YES', 'NO, rerun')
head(NOXchk)
```

```
##      Run_Info      Sample_Name      Conc      Abs Dilution
## 65 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04 0.717276 0.742243      0
## 68 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04 0.722862 0.746443      0
## 261 RESULT peCheck_N03_706 0.711677 0.726175      0
## 262 RESULT peCheck_N03_706 0.684810 0.705487      0
## 439 RESULT peCheck_N03_706ppm_A 0.671314 0.714344      0
## 440 RESULT peCheck_N03_706ppm_B 0.673346 0.715908      0
##      Units      Test Run_Number ConcTrue Chk_diff Chk_dff_flag
## 65 mg N/L Vanadium NOx          1      0.706 1.5845135          YES
## 68 mg N/L Vanadium NOx          1      0.706 2.3601999          YES
## 261 mg N/L Vanadium NOx          2      0.706 0.8008876          YES
## 262 mg N/L Vanadium NOx          2      0.706 -3.0471452          YES
## 439 mg N/L Vanadium NOx          3      0.706 -5.0367600          YES
## 440 mg N/L Vanadium NOx          3      0.706 -4.7347076          YES
```

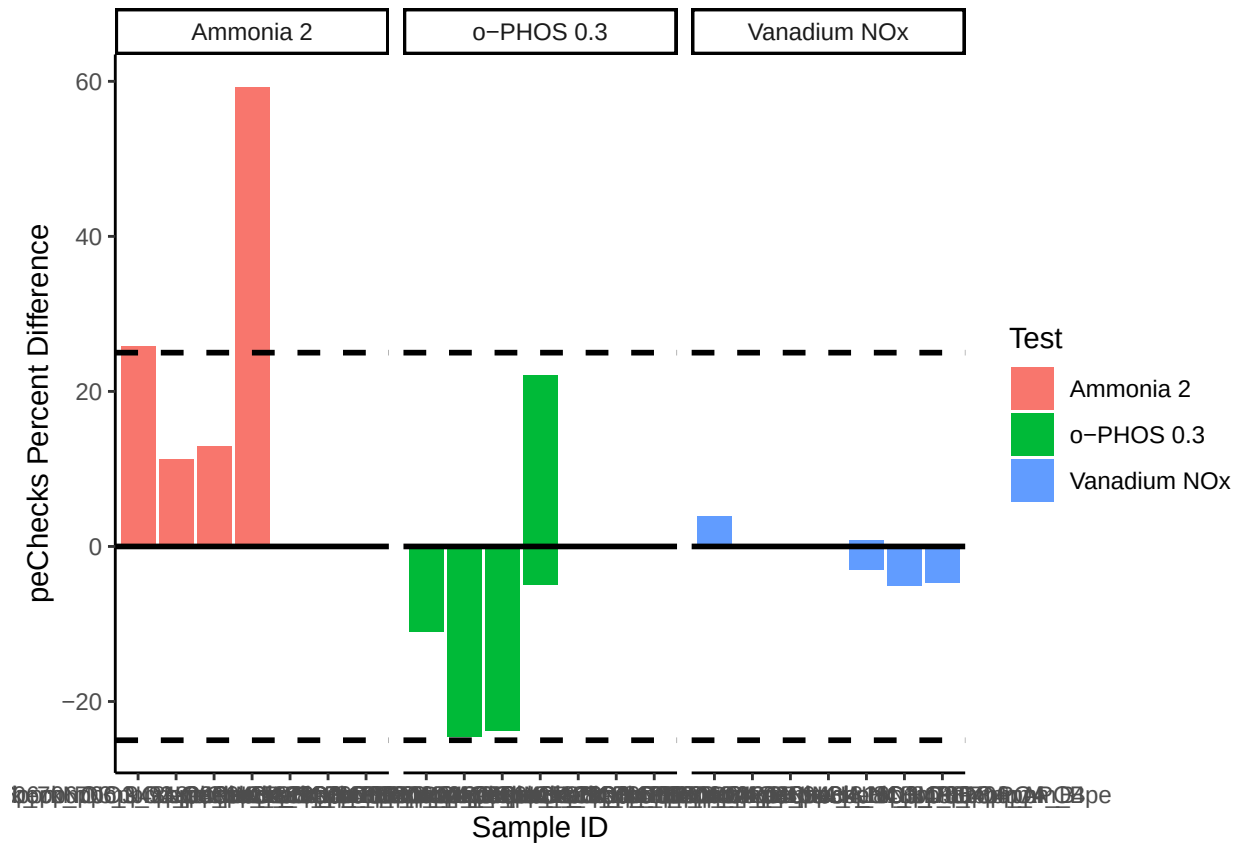
```
Allchks <- rbind(NH4chk, PO4chk)
Allchks <- rbind(Allchks, NOXchk)
head(Allchks)
```

```
##      Run_Info      Sample_Name      Conc      Abs
## 142  RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 1.285274 0.762447
## 149  RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 1.270126 0.753577
## 571  RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04 1.229047 0.750897
## 1060 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04_A 1.061506 0.682024
## 1061 RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04_B 1.079544 0.693616
## 327  RESULT peCheck_706ppmN03_948ppmNH4_818ppmP04pe 0.796170 0.326971
##      Dilution  Units      Test Run_Number ConcTrue  Chk_diff Chk_dff_flag
## 142      0 mg N/L  Ammonia 2          1      0.948 30.204444      NO, rerun
## 149      0 mg N/L  Ammonia 2          1      0.948 29.044878      NO, rerun
## 571      0 mg N/L  Ammonia 2          2      0.948 25.819103      NO, rerun
## 1060      0 mg N/L  Ammonia 2          3      0.948 11.296906          YES
## 1061      0 mg N/L  Ammonia 2          3      0.948 12.975699          YES
## 327      0 mg P/L  o-PHOS 0.3          1      0.818 -2.704796          YES
```

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Chksbar <- ggplot(data = Allchks, aes(x = Sample_Name, y = Chk_diff, fill=Test)) +
  geom_bar(stat = 'identity') +
  facet_wrap(~Test) +
  theme_classic() + labs(x= "Sample ID", y="peChecks Percent Difference") +
  theme(legend.position="right") +
  geom_hline(yintercept=25, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=0, color = "black", size=1) +
  geom_hline(yintercept=-25, linetype="dashed", color = "black", size=1)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
Chksbar
```



Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## Duplicated samples: MSM_UP_LysC_20cm, MSM_WC_SipC_45cm, MSM_UP_LysB_10cm, MSM_UP_LysA_20cm, MSM_UP_L
```

```
##
```

```
## WARNING: Duplicated samples with NO dilution (possible input error):
```

```
## # A tibble: 3 x 4
```

```
##   Test      Sample_Name    n_dups any_dilution
##   <chr>      <chr>          <int> <lgl>
## 1 o-PHOS 0.3 GWI_TR_RHZ_SF_8      2 FALSE
## 2 o-PHOS 0.3 GWI_WC_SipA_20cm     2 FALSE
## 3 o-PHOS 0.3 GWI_WC_SipC_45cm     2 FALSE
```

NOx Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## No duplicated samples.
```

Constants

```
N_mw <- 14.0067 # molecular weight of N
P_mw <- 30.973762 # molecular weight of P
Con1 <- 1000 # conversion factor value
Con2 <- 1000000 # conversion factor value
```

Convert Data from mg/L to uM

```
head(alldat2)
```

```
## # A tibble: 6 x 10
##   Run_Info Sample_Name      Conc  Abs  Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>      <chr> <chr>    <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_UP_LysB_10cm 0.042~ 0.03~      1 mg N~ Ammo~ 2      MSM_UP~
## 2 RESULT    MSM_UP_LysC_10cm 0.568~ 0.35~      1 mg N~ Ammo~ 2      MSM_UP~
## 3 RESULT    MSM_UP_LysA_20cm 1.795~ 1.09~      1 mg N~ Ammo~ 2      MSM_UP~
## 4 RESULT    MSM_UP_LysB_20cm 0.588~ 0.36~      1 mg N~ Ammo~ 2      MSM_UP~
## 5 RESULT    MSM_UP_LysA_45cm 1.148~ 0.70~      1 mg N~ Ammo~ 2      MSM_UP~
## 6 RESULT    MSM_UP_LysB_45cm 0.911~ 0.55~      1 mg N~ Ammo~ 2      MSM_UP~
## # i 1 more variable: Original_Name <chr>
```

```
head(Nalldat2)
```

```
## # A tibble: 6 x 10
##   Run_Info Sample_Name      Conc  Abs  Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>      <dbl> <dbl>    <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_UP_LysB_1~ 1.10e-3 0.163      1 mg N~ Vana~ 2      MSM_UP~
## 2 RESULT    MSM_UP_LysC_1~ 2.49e-3 0.164      1 mg N~ Vana~ 2      MSM_UP~
## 3 RESULT    MSM_UP_LysA_2~ 1.26e-3 0.163      1 mg N~ Vana~ 2      MSM_UP~
## 4 RESULT    MSM_UP_LysB_2~ 3.20e-3 0.164      1 mg N~ Vana~ 2      MSM_UP~
## 5 RESULT    MSM_UP_LysC_2~ -1.92e-3 0.160      1 mg N~ Vana~ 2      MSM_UP~
## 6 RESULT    MSM_UP_LysA_4~ 6.60e-5 0.162      1 mg N~ Vana~ 2      MSM_UP~
## # i 1 more variable: Original_Name <chr>
```

```
#subset by test
```

```
NH4samples <- subset(alldat2, Test == "Ammonia 2")
```

```
head(NH4samples)
```

```
## # A tibble: 6 x 10
##   Run_Info Sample_Name      Conc  Abs  Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>      <chr> <chr>    <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_UP_LysB_10cm 0.042~ 0.03~      1 mg N~ Ammo~ 2      MSM_UP~
## 2 RESULT    MSM_UP_LysC_10cm 0.568~ 0.35~      1 mg N~ Ammo~ 2      MSM_UP~
## 3 RESULT    MSM_UP_LysA_20cm 1.795~ 1.09~      1 mg N~ Ammo~ 2      MSM_UP~
## 4 RESULT    MSM_UP_LysB_20cm 0.588~ 0.36~      1 mg N~ Ammo~ 2      MSM_UP~
```



```
## 5 RESULT MSM_UP_LysA_45cm 1.148~ 0.70~ 1 mg N~ Ammo~ 2 MSM_UP~
## 6 RESULT MSM_UP_LysB_45cm 0.911~ 0.55~ 1 mg N~ Ammo~ 2 MSM_UP~
## # i 1 more variable: Original_Name <chr>
```

```
P04samples <- subset(allldat2, Test == "o-PHOS 0.3")
head(P04samples)
```

```
## # A tibble: 6 x 10
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
## <chr> <chr> <chr> <chr> <dbl> <chr> <chr> <chr> <chr>
## 1 RESULT MSM_UP_LysC_10cm 0.122~ 0.05~ 1 mg P~ o-PH~ 2 MSM_UP~
## 2 RESULT MSM_UP_LysB_20cm 0.099~ 0.04~ 1 mg P~ o-PH~ 2 MSM_UP~
## 3 RESULT MSM_UP_LysC_20cm 0.155~ 0.06~ 1 mg P~ o-PH~ 2 MSM_UP~
## 4 RESULT MSM_UP_LysB_45cm 0.175~ 0.07~ 1 mg P~ o-PH~ 2 MSM_UP~
## 5 RESULT MSM_UP_LysC_45cm 0.159~ 0.06~ 1 mg P~ o-PH~ 2 MSM_UP~
## 6 RESULT MSM_TR_LysB_10cm 0.083~ 0.03~ 1 mg P~ o-PH~ 2 MSM_TR~
## # i 1 more variable: Original_Name <chr>
```

```
NOXsamples <- subset(Nalldat2, Test == "Vanadium NOx")
head(NOXsamples)
```

```
## # A tibble: 6 x 10
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
## <chr> <chr> <dbl> <dbl> <dbl> <chr> <chr> <chr> <chr>
## 1 RESULT MSM_UP_LysB_1~ 1.10e-3 0.163 1 mg N~ Vana~ 2 MSM_UP~
## 2 RESULT MSM_UP_LysC_1~ 2.49e-3 0.164 1 mg N~ Vana~ 2 MSM_UP~
## 3 RESULT MSM_UP_LysA_2~ 1.26e-3 0.163 1 mg N~ Vana~ 2 MSM_UP~
## 4 RESULT MSM_UP_LysB_2~ 3.20e-3 0.164 1 mg N~ Vana~ 2 MSM_UP~
## 5 RESULT MSM_UP_LysC_2~ -1.92e-3 0.160 1 mg N~ Vana~ 2 MSM_UP~
## 6 RESULT MSM_UP_LysA_4~ 6.60e-5 0.162 1 mg N~ Vana~ 2 MSM_UP~
## # i 1 more variable: Original_Name <chr>
```

```
NH4samples$Conc_uM <- (((as.numeric(NH4samples$Conc))/Con1)/N_mw)*Con2
head(NH4samples)
```

```
## # A tibble: 6 x 11
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
## <chr> <chr> <chr> <chr> <dbl> <chr> <chr> <chr> <chr>
## 1 RESULT MSM_UP_LysB_10cm 0.042~ 0.03~ 1 mg N~ Ammo~ 2 MSM_UP~
## 2 RESULT MSM_UP_LysC_10cm 0.568~ 0.35~ 1 mg N~ Ammo~ 2 MSM_UP~
## 3 RESULT MSM_UP_LysA_20cm 1.795~ 1.09~ 1 mg N~ Ammo~ 2 MSM_UP~
## 4 RESULT MSM_UP_LysB_20cm 0.588~ 0.36~ 1 mg N~ Ammo~ 2 MSM_UP~
## 5 RESULT MSM_UP_LysA_45cm 1.148~ 0.70~ 1 mg N~ Ammo~ 2 MSM_UP~
## 6 RESULT MSM_UP_LysB_45cm 0.911~ 0.55~ 1 mg N~ Ammo~ 2 MSM_UP~
## # i 2 more variables: Original_Name <chr>, Conc_uM <dbl>
```

```
P04samples$Conc_uM <- (((as.numeric(P04samples$Conc))/Con1)/N_mw)*Con2
head(P04samples)
```

```
## # A tibble: 6 x 11
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
```

```
##   <chr>      <chr>          <chr> <chr>      <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_UP_LysC_10cm 0.122~ 0.05~      1 mg P~ o-PH~ 2      MSM_UP~
## 2 RESULT    MSM_UP_LysB_20cm 0.099~ 0.04~      1 mg P~ o-PH~ 2      MSM_UP~
## 3 RESULT    MSM_UP_LysC_20cm 0.155~ 0.06~      1 mg P~ o-PH~ 2      MSM_UP~
## 4 RESULT    MSM_UP_LysB_45cm 0.175~ 0.07~      1 mg P~ o-PH~ 2      MSM_UP~
## 5 RESULT    MSM_UP_LysC_45cm 0.159~ 0.06~      1 mg P~ o-PH~ 2      MSM_UP~
## 6 RESULT    MSM_TR_LysB_10cm 0.083~ 0.03~      1 mg P~ o-PH~ 2      MSM_TR~
## # i 2 more variables: Original_Name <chr>, Conc_uM <dbl>
```

```
NOXsamples$Conc_uM_raw <- (((as.numeric(NOXsamples$Conc))/Con1)/N_mw)*Con2
head(NOXsamples)
```

```
## # A tibble: 6 x 11
##   Run_Info Sample_Name      Conc  Abs Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>          <dbl> <dbl>      <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_UP_LysB_1~  1.10e-3 0.163      1 mg N~ Vana~ 2      MSM_UP~
## 2 RESULT    MSM_UP_LysC_1~  2.49e-3 0.164      1 mg N~ Vana~ 2      MSM_UP~
## 3 RESULT    MSM_UP_LysA_2~  1.26e-3 0.163      1 mg N~ Vana~ 2      MSM_UP~
## 4 RESULT    MSM_UP_LysB_2~  3.20e-3 0.164      1 mg N~ Vana~ 2      MSM_UP~
## 5 RESULT    MSM_UP_LysC_2~ -1.92e-3 0.160      1 mg N~ Vana~ 2      MSM_UP~
## 6 RESULT    MSM_UP_LysA_4~  6.60e-5 0.162      1 mg N~ Vana~ 2      MSM_UP~
## # i 2 more variables: Original_Name <chr>, Conc_uM_raw <dbl>
```

```
#add step to make negative values equal to bd (below detection) and replace with zeros
NOXsamples$Conc_uM <- ifelse(NOXsamples$Conc_uM_raw<0, 0, (NOXsamples$Conc_uM_raw) )
head(NOXsamples)
```

```
## # A tibble: 6 x 12
##   Run_Info Sample_Name      Conc  Abs Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>          <dbl> <dbl>      <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_UP_LysB_1~  1.10e-3 0.163      1 mg N~ Vana~ 2      MSM_UP~
## 2 RESULT    MSM_UP_LysC_1~  2.49e-3 0.164      1 mg N~ Vana~ 2      MSM_UP~
## 3 RESULT    MSM_UP_LysA_2~  1.26e-3 0.163      1 mg N~ Vana~ 2      MSM_UP~
## 4 RESULT    MSM_UP_LysB_2~  3.20e-3 0.164      1 mg N~ Vana~ 2      MSM_UP~
## 5 RESULT    MSM_UP_LysC_2~ -1.92e-3 0.160      1 mg N~ Vana~ 2      MSM_UP~
## 6 RESULT    MSM_UP_LysA_4~  6.60e-5 0.162      1 mg N~ Vana~ 2      MSM_UP~
## # i 3 more variables: Original_Name <chr>, Conc_uM_raw <dbl>, Conc_uM <dbl>
```

Pull all data back together and add flags

```
#pull out the columns we want from each dataframe
NH4_pull <- NH4samples[ ,c(2,3,11) ]
head(NH4_pull)
```

```
## # A tibble: 6 x 3
##   Sample_Name      Conc      Conc_uM
##   <chr>          <chr>      <dbl>
## 1 MSM_UP_LysB_10cm 0.042758    3.05
## 2 MSM_UP_LysC_10cm 0.568162   40.6
## 3 MSM_UP_LysA_20cm 1.795753  128.
```

```
## 4 MSM_UP_LysB_20cm 0.588825 42.0
## 5 MSM_UP_LysA_45cm 1.148694 82.0
## 6 MSM_UP_LysB_45cm 0.911524 65.1
```

```
P04_pull <- P04samples[ ,c(2,3,11) ]
head(P04_pull)
```

```
## # A tibble: 6 x 3
##   Sample_Name      Conc      Conc_uM
##   <chr>          <chr>    <dbl>
## 1 MSM_UP_LysC_10cm 0.122153    8.72
## 2 MSM_UP_LysB_20cm 0.099871    7.13
## 3 MSM_UP_LysC_20cm 0.155479   11.1
## 4 MSM_UP_LysB_45cm 0.175185   12.5
## 5 MSM_UP_LysC_45cm 0.159733   11.4
## 6 MSM_TR_LysB_10cm 0.083248    5.94
```

```
NOX_pull <- NOXsamples[ ,c(2,3,12) ]
head(NOX_pull)
```

```
## # A tibble: 6 x 3
##   Sample_Name      Conc Conc_uM
##   <chr>          <dbl>  <dbl>
## 1 MSM_UP_LysB_10cm 0.00110 0.0787
## 2 MSM_UP_LysC_10cm 0.00249 0.178
## 3 MSM_UP_LysA_20cm 0.00126 0.0901
## 4 MSM_UP_LysB_20cm 0.00320 0.228
## 5 MSM_UP_LysC_20cm -0.00192 0
## 6 MSM_UP_LysA_45cm 0.000066 0.00471
```

#Bring all this data back together:

```
all_data <- merge(NH4_pull, P04_pull, by="Sample_Name", all.x=TRUE)
all_data <- merge(all_data, NOX_pull, by="Sample_Name", all.x=TRUE)
head(all_data)
```

```
##           Sample_Name      Conc.x Conc_uM.x      Conc.y Conc_uM.y      Conc
## 1 GCrew_TR_LysA_20cm 0.306868 21.908658 0.038776 2.768389 0.000036
## 2 GCrew_TR_LysA_45cm 0.073945 5.279259 0.025919 1.850472 -0.004228
## 3 GCrew_TR_LysB_10cm 1.393605 99.495599 0.067836 4.843111 0.001133
## 4 GCrew_TR_LysB_20cm 1.047486 74.784639 0.036769 2.625101 -0.000479
## 5 GCrew_TR_LysB_45cm 0.04421 3.156347 0.031594 2.255635 0.008479
## 6 GCrew_TR_LysC_10cm 0.042944 3.065961 0.034448 2.459394 -0.007843
##           Conc_uM
## 1 0.002570199
## 2 0.000000000
## 3 0.080889860
## 4 0.000000000
## 5 0.605353152
## 6 0.000000000
```

```
colnames(all_data) <- c("Sample_Name", "NH3_mgL", "NH3_uM", "PO4_mgL", "PO4_uM", "NOx_mgL", "NOx_uM")
head(all_data)
```

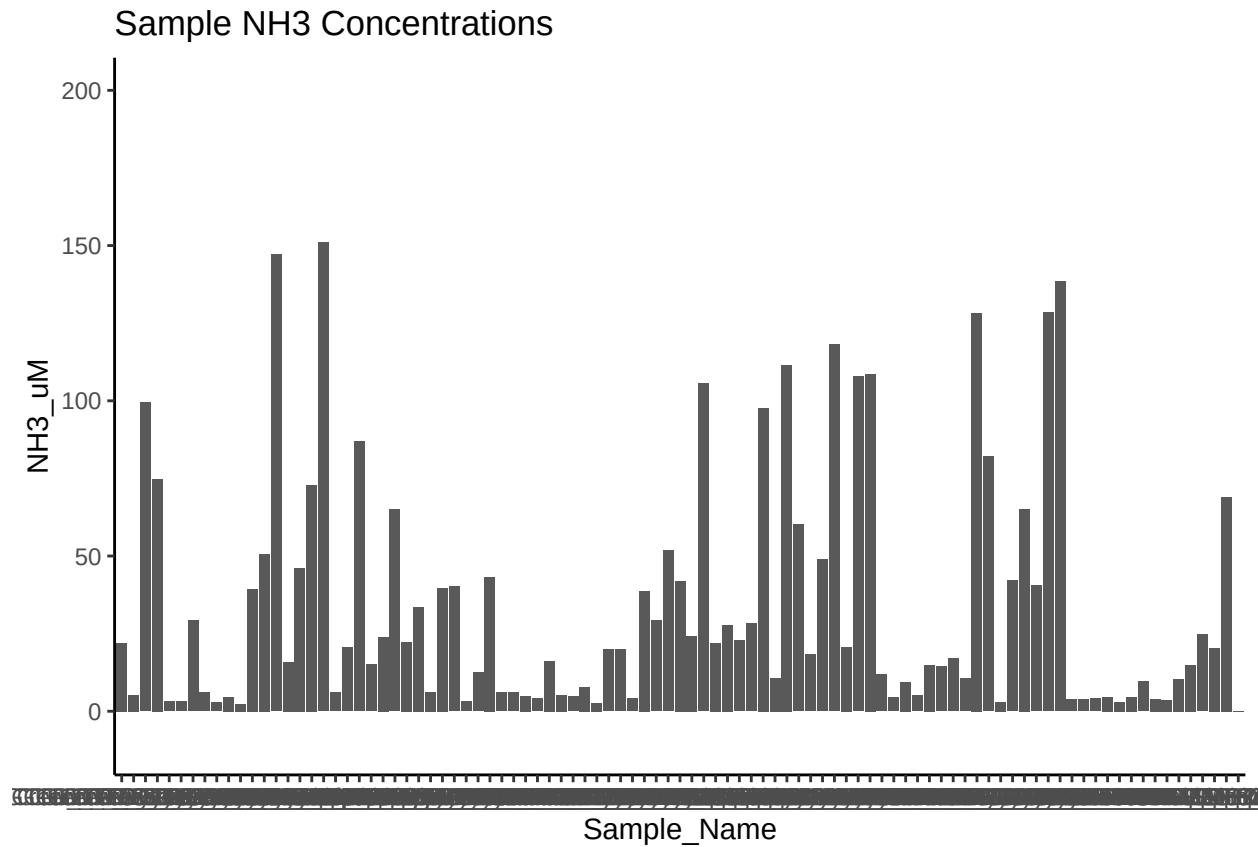
```
##           Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL  NOx_uM
## 1 GCrew_TR_LysA_20cm 0.306868 21.908658 0.038776 2.768389 0.000036 0.002570199
## 2 GCrew_TR_LysA_45cm 0.073945 5.279259 0.025919 1.850472 -0.004228 0.000000000
## 3 GCrew_TR_LysB_10cm 1.393605 99.495599 0.067836 4.843111 0.001133 0.080889860
## 4 GCrew_TR_LysB_20cm 1.047486 74.784639 0.036769 2.625101 -0.000479 0.000000000
## 5 GCrew_TR_LysB_45cm 0.04421 3.156347 0.031594 2.255635 0.008479 0.605353152
## 6 GCrew_TR_LysC_10cm 0.042944 3.065961 0.034448 2.459394 -0.007843 0.000000000
```

```
#add in an if then statement that tells us if they are within the range of the test - check this after
all_data$NH3_range <- ifelse(all_data$NH3_mgL<0.02, "bd1", ifelse(all_data$NH3_mgL>2, "ad1", "Within_Range"))
all_data$PO4_range <- ifelse(all_data$PO4_mgL<0.003, "bd1", ifelse(all_data$PO4_mgL>3, "ad1", "Within_Range"))
all_data$NOx_range <- ifelse(all_data$NOx_mgL<0.025, "bd1", ifelse(all_data$NOx_mgL>1, "ad1", "Within_Range"))
head(all_data)
```

```
##           Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL  NOx_uM
## 1 GCrew_TR_LysA_20cm 0.306868 21.908658 0.038776 2.768389 0.000036 0.002570199
## 2 GCrew_TR_LysA_45cm 0.073945 5.279259 0.025919 1.850472 -0.004228 0.000000000
## 3 GCrew_TR_LysB_10cm 1.393605 99.495599 0.067836 4.843111 0.001133 0.080889860
## 4 GCrew_TR_LysB_20cm 1.047486 74.784639 0.036769 2.625101 -0.000479 0.000000000
## 5 GCrew_TR_LysB_45cm 0.04421 3.156347 0.031594 2.255635 0.008479 0.605353152
## 6 GCrew_TR_LysC_10cm 0.042944 3.065961 0.034448 2.459394 -0.007843 0.000000000
##           NH3_range  PO4_range NOx_range
## 1 Within_Range Within_Range      bd1
## 2 Within_Range Within_Range      bd1
## 3 Within_Range Within_Range      bd1
## 4 Within_Range Within_Range      bd1
## 5 Within_Range Within_Range      bd1
## 6 Within_Range Within_Range      bd1
```

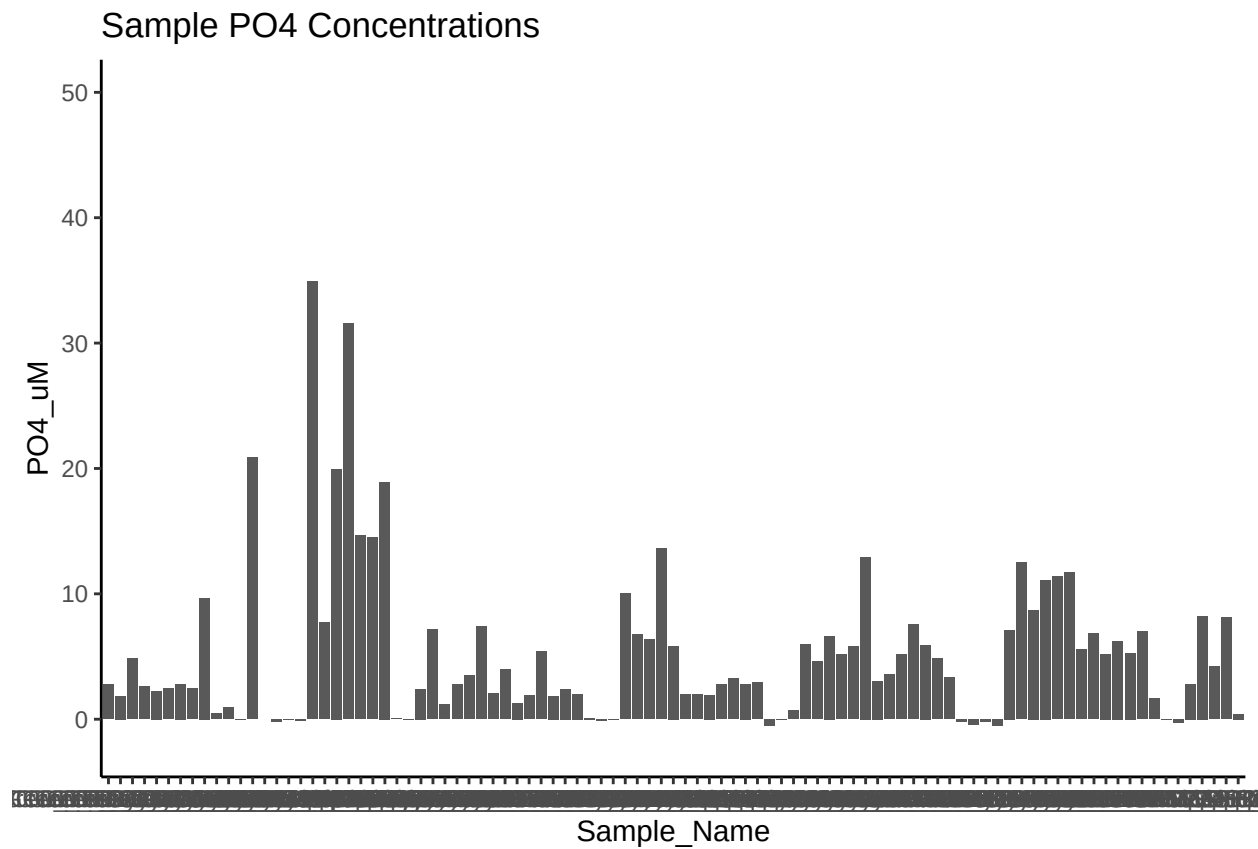
Take an initial look at concentrations

```
#plot data to get a sense of any outliers
NH3look <- ggplot(data=all_data, aes(x=Sample_Name, y=NH3_uM)) +
  geom_bar(stat="identity") +
  theme_classic() + ylim(-10, 200) +
  theme(legend.position="none") +
  ggtitle("Sample NH3 Concentrations")
NH3look
```

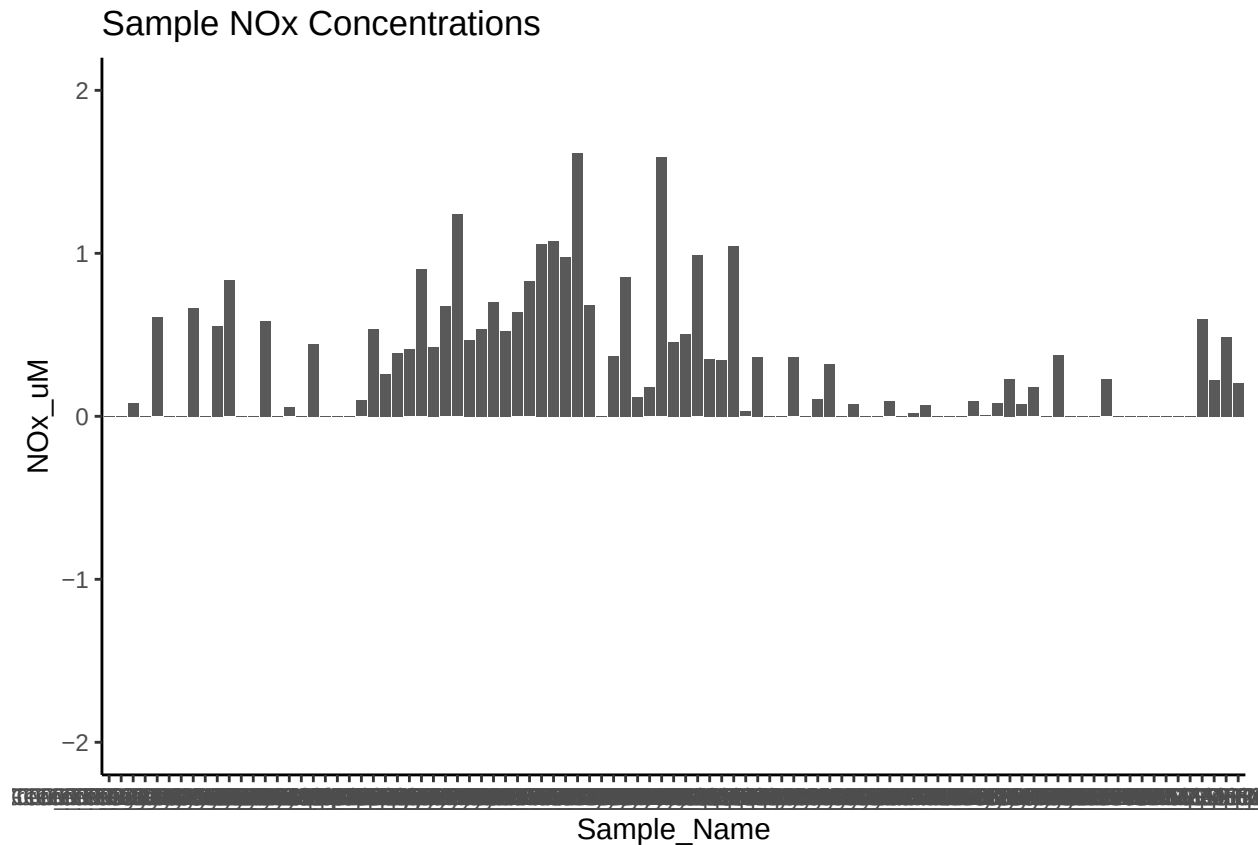


```
P04look <- ggplot(data=all_data, aes(x=Sample_Name, y=P04_uM)) +
  geom_bar(stat="identity") +
  theme_classic() + ylim(-2, 50) +
  theme(legend.position="none") +
  ggtitle("Sample P04 Concentrations")
P04look
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



```
NOXlook <- ggplot(data=all_data, aes(x=Sample_Name, y=NOx_uM)) +
  geom_bar(stat="identity") +
  theme_classic() + ylim(-2, 2) +
  theme(legend.position="none") +
  ggtitle("Sample NOx Concentrations")
NOXlook
```



Pull out data you need, make IDs

```
head(all_data)
```

```
##           Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL  NOx_uM
## 1 GCrew_TR_LysA_20cm 0.306868 21.908658 0.038776 2.768389 0.000036 0.002570199
## 2 GCrew_TR_LysA_45cm 0.073945  5.279259 0.025919 1.850472 -0.004228 0.000000000
## 3 GCrew_TR_LysB_10cm 1.393605 99.495599 0.067836 4.843111 0.001133 0.080889860
## 4 GCrew_TR_LysB_20cm 1.047486 74.784639 0.036769 2.625101 -0.000479 0.000000000
## 5 GCrew_TR_LysB_45cm 0.04421  3.156347 0.031594 2.255635 0.008479 0.605353152
## 6 GCrew_TR_LysC_10cm 0.042944 3.065961 0.034448 2.459394 -0.007843 0.000000000
##      NH3_range  PO4_range NOx_range
## 1 Within_Range Within_Range      bdl
## 2 Within_Range Within_Range      bdl
## 3 Within_Range Within_Range      bdl
## 4 Within_Range Within_Range      bdl
## 5 Within_Range Within_Range      bdl
## 6 Within_Range Within_Range      bdl
```

```
# out <- all_data[,c(1,3,5,7,8,9,10)]
out <- all_data
head(out)
```

```
##      Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL  NOx_uM
## 1 GCrew_TR_LysA_20cm 0.306868 21.908658 0.038776 2.768389 0.000036 0.002570199
## 2 GCrew_TR_LysA_45cm 0.073945 5.279259 0.025919 1.850472 -0.004228 0.000000000
## 3 GCrew_TR_LysB_10cm 1.393605 99.495599 0.067836 4.843111 0.001133 0.080889860
## 4 GCrew_TR_LysB_20cm 1.047486 74.784639 0.036769 2.625101 -0.000479 0.000000000
## 5 GCrew_TR_LysB_45cm 0.04421 3.156347 0.031594 2.255635 0.008479 0.605353152
## 6 GCrew_TR_LysC_10cm 0.042944 3.065961 0.034448 2.459394 -0.007843 0.000000000
##      NH3_range  PO4_range NOx_range
## 1 Within_Range Within_Range      bdl
## 2 Within_Range Within_Range      bdl
## 3 Within_Range Within_Range      bdl
## 4 Within_Range Within_Range      bdl
## 5 Within_Range Within_Range      bdl
## 6 Within_Range Within_Range      bdl
```

```
#for steph <- pull out identifiers of the sample names
#pull the sample ID and separate it by the underscores
IDs <- data.frame(do.call('rbind', strsplit(as.character(out$Sample_Name), '_', fixed=TRUE)))
```

```
## Warning in rbind(c("GCrew", "TR", "LysA", "20cm"), c("GCrew", "TR", "LysA", :
## number of columns of result is not a multiple of vector length (arg 1)
```

```
colnames(IDs) <- c("Site", "Zone", "Replicate", "Depth", "Tree")
IDs$Date <- 202207
IDs$Month <- "July"
head(IDs)
```

```
##      Site Zone Replicate Depth  Tree  Date Month
## 1 GCrew  TR      LysA   20cm GCrew 202207  July
## 2 GCrew  TR      LysA   45cm GCrew 202207  July
## 3 GCrew  TR      LysB   10cm GCrew 202207  July
## 4 GCrew  TR      LysB   20cm GCrew 202207  July
## 5 GCrew  TR      LysB   45cm GCrew 202207  July
## 6 GCrew  TR      LysC   10cm GCrew 202207  July
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, out)
head(alldat)
```

```
##      Site Zone Replicate Depth  Tree  Date Month      Sample_Name NH3_mgL
## 1 GCrew  TR      LysA   20cm GCrew 202207  July GCrew_TR_LysA_20cm 0.306868
## 2 GCrew  TR      LysA   45cm GCrew 202207  July GCrew_TR_LysA_45cm 0.073945
## 3 GCrew  TR      LysB   10cm GCrew 202207  July GCrew_TR_LysB_10cm 1.393605
## 4 GCrew  TR      LysB   20cm GCrew 202207  July GCrew_TR_LysB_20cm 1.047486
## 5 GCrew  TR      LysB   45cm GCrew 202207  July GCrew_TR_LysB_45cm 0.04421
## 6 GCrew  TR      LysC   10cm GCrew 202207  July GCrew_TR_LysC_10cm 0.042944
##      NH3_uM  PO4_mgL  PO4_uM  NOx_mgL  NOx_uM  NH3_range  PO4_range
## 1 21.908658 0.038776 2.768389 0.000036 0.002570199 Within_Range Within_Range
## 2 5.279259 0.025919 1.850472 -0.004228 0.000000000 Within_Range Within_Range
## 3 99.495599 0.067836 4.843111 0.001133 0.080889860 Within_Range Within_Range
## 4 74.784639 0.036769 2.625101 -0.000479 0.000000000 Within_Range Within_Range
## 5 3.156347 0.031594 2.255635 0.008479 0.605353152 Within_Range Within_Range
```



```
## 6  3.065961 0.034448 2.459394 -0.007843 0.000000000 Within_Range Within_Range
##   NOx_range
## 1      bdl
## 2      bdl
## 3      bdl
## 4      bdl
## 5      bdl
## 6      bdl
```

```
alldat <- alldat %>%
  filter(!Sample_Name == "GCrew_WC_SipA_45cm_2")
```

Export final data with flags

```
#Export Data
#setwd("S:/Biogeochemistry/People/Wilson (Steph)/Data/SEAL/Final Data Files")

alldat$Run_notes = Run_notes

write.csv(alldat, file="Processed Data/COMPASS_SynopticCB_Nutrients_202207.csv")
```